

Exact Radial-Determinant Frequencies of a Piezoelectric Annular Plate, and Inverse Recovery of e_{31} from Open-Circuit Flexure

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Abstract

A steel annular plate with identical thickness-polarized piezoceramic layers on both faces is solved by exact radial wave functions and a boundary determinant. Under short-circuit electrodes a through-thickness potential that vanishes on every electrode, projected by the same Galerkin weight as the flexural equation, reduces the coupled flexural problem to three Helmholtz branches and a 6×6 edge matrix. That construction recovers all 27 non-trivial frequencies of Duan, Quek and Wang’s Table 4: every root rounds to the printed integer, and the worst relative error is -0.016% . The same short-circuit flexural frequency is nonetheless a poor observable for the piezoelectric stress constant e_{31} : a 0.01% error in frequency maps to a 330% error in recovered e_{31} , while the host Young’s modulus E recovers at 0.025% . The obstruction is electrical. Fully electroded Kirchhoff flexure with a spatially constant electrode voltage produces a constant piezo moment; Green’s identity then forces the net electrode charge to vanish whenever the edge rotation is zero, so clamped-clamped open-circuit coincides with short-circuit at leading order. Free-free axisymmetric open-circuit does not. A linear through-thickness potential with $Q = 0$ on the outer-electrode bus updates the elastic 4×4 by a rank-1 moment term. With the bus charge taken as the variable conjugate to V – so that the charge and moment rows share the same lever arm and the operator is reciprocal – it raises the first $n = 0$ root by 1.950% , 2.776% , and 4.016% at $h_1/2h = 1/12$, $1/8$, and $1/5$. The same update on a clamped-inner, free-outer (C-F) pair raises that root by 1.826% at $h_1/2h = 1/12$; the swapped pair (F-C) splits by only 0.039% , because only the small inner edge sees the piezo moment. Axisymmetric piezoelectric finite elements on a fair both-faces-grounded partner reproduce the split at all three ratios to $99.5\text{--}99.7\%$ of its value, and with only the inner radius clamped reproduce the clamped-free split to within 0.9% . From those open-circuit frequencies, 0.01% frequency precision maps to 0.87% , 0.63% , and 0.45% uncertainty in e_{31} when the host modulus is known exactly; recovering E and e_{31} in the sequence an experiment actually would compounds the $h_1/2h = 1/12$ figure to 1.21% . The driven admittance $Y(\omega) = j\omega Q/V$ of the same model has its pole at the short-circuit root and its zero at the open-circuit root; an independent bisection of Q/V reproduces the open-circuit frequency to better than 10^{-13} relative at each of the three Table 4 ratios, and $k_{\text{eff}}^2 = 1 - (f_r/f_a)^2 = 0.0379$ at $h_1/2h = 1/12$.

Keywords: piezoelectric annular plate; open-circuit; short-circuit; inverse parameter estimation; electromechanical coupling; radial determinant; free-free flexure; clamped-free.

1 Introduction

Piezoelectric annular plates are used as transducers, sensors, and frequency-control elements, and the constants that govern them are identified in practice from electrical resonance and antiresonance [7]. The corresponding free-vibration analysis is usually approximate: finite elements, Rayleigh–Ritz, or first-order shear deformation. Exact two-dimensional reductions exist, but they are almost always posed under short-circuit electrodes. Duan, Quek and Wang [5] coupled Kirchhoff and Mindlin annular plates to thickness-polarized piezoceramic layers with a sinusoidal through-thickness potential that vanishes on both electrode faces, and tabulated clamped–clamped frequencies. Liu, Wang and Quek [6] gave the circular Mindlin analogue (Supplementary §S.10 checks our Kirchhoff solid-disk roots against their classical-plate column: eleven of twelve roots round to their printed values, but changing e_{31} from its printed value to nearly zero moves those roots by less than 10^{-4} , so that check covers mainly the laminate reduction). In both constructions the electrodes are shorted, and Duan, Quek and Wang’s own decomposition of the frequency shift (their Fig. 3) already shows that, under that electrical condition, ordinary added-layer stiffness dominates electromechanical coupling by two to three orders of magnitude. Hosseini-Hashemi, Kalbasi and Rokni Damavandi Taher treated piezoelectric annuli of variable thickness [8]; Parashar gave a Rayleigh–Ritz annular actuator with experiment [9]. None of those papers inverts a flexural frequency for e_{31} , and none replaces the short-circuit potential by an open-circuit electrode bus on the same exact radial operator.

Laboratory identification of piezoelectric constants follows IEEE Std 176 [7]: resonance and antiresonance of an electrically driven specimen, not a list of short-circuit flexural eigenvalues. The gap between those two practices is the subject of this paper.

That gap matters as soon as the same frequencies are used to identify e_{31} . A short-circuit flexural root is an excellent thermometer for the host modulus and a nearly useless thermometer for the piezoelectric stress constant. Closing the gap requires an electrical boundary condition in which the electrode voltage can do mechanical work, and a mechanical boundary condition that sees the resulting moment. The present paper supplies both, as exact radial determinants, and then inverts them.

The underlying elastic operator is the closed-annulus reduction of the Seok–Tiersten–Scarton wave-function method [1, 2, 3, 4]: integer circumferential order n , a 4×4 radial edge matrix, and frequencies as zeros of that determinant. That elastic ring has been checked against Vogel and Skinner, Narita, Southwell, and Irie by McCune and Stutts [12], who also inverted Southwell’s geometric ratio b/a at fixed frequency by the same root-search and on-branch sensitivity recipe used here. The present work does not re-derive that elastic baseline. It adds piezoelectric layers, two electrical boundary conditions, a literature-table validation, a finite-element open-circuit check, a quantified inverse, and the driven admittance whose poles and zeros *are* the short-circuit and open-circuit roots.

What is not claimed: in-plane (extensional) piezoelectric coupling; a monolithic piezoceramic ring; simultaneous multi-parameter recovery from one mode; or a laboratory specimen. The last of those would use the free–free $n = 0$ resonance/antiresonance pair derived below and is left as an experiment, not as a missing numerical result.

2 Layered annulus

The host occupies $|z| < h$ and is isotropic, with Young’s modulus E , Poisson ratio ν , and density ρ . Identical thickness-polarized piezoceramic layers occupy $h \leq z \leq h + h_1$ and $-h - h_1 \leq z \leq -h$,

with stiffnesses c_{ij}^E at constant electric field, stress constants e_{31} , e_{33} , e_{15} , permittivities Ξ_{11} , Ξ_{33} , and density ρ_p . Numerical comparisons use the steel/PZT-4 constants of Duan, Quek and Wang [5]: $r_i = 0.1$ m, $r_o = 0.6$ m, $h = 0.01$ m, $E = 200$ GPa, $\nu = 0.3$, $\rho = 7800$ kg m $^{-3}$, $c_{11}^E = 132$, $c_{12}^E = 71$, $c_{13}^E = 73$, $c_{33}^E = 115$ GPa, $e_{31} = -4.1$, $e_{33} = 14.1$ C m $^{-2}$, $\Xi_{11} = 7.124 \times 10^{-9}$, $\Xi_{33} = 5.841 \times 10^{-9}$ F m $^{-1}$, $\rho_p = 7500$ kg m $^{-3}$. Thickness ratios $h_1/2h = 1/12$, $1/8$, and $1/5$ are those of their Table 4. The constants are those printed in their Table 1 and in Liu, Wang and Quek's Table 1 [6], including the sign of e_{31} . Figure 1 is a through-thickness slice: inner electrodes at the host/piezo interfaces, outer electrodes on the free faces.

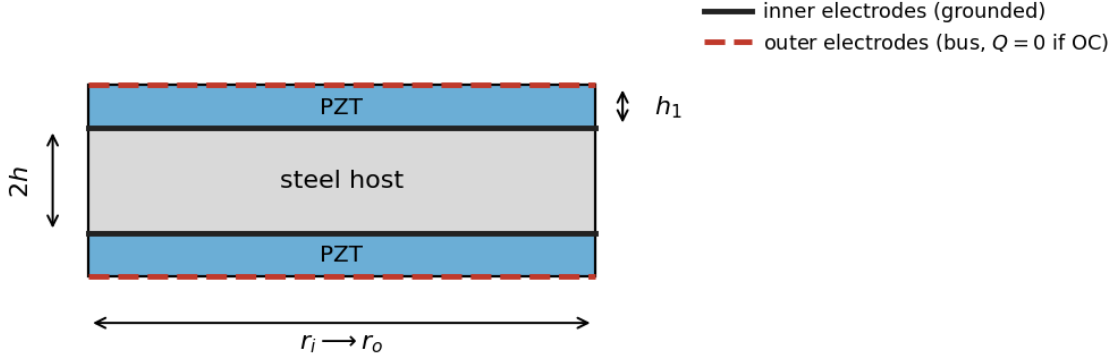


Figure 1: Radial-thickness slice of the layered annulus (not to scale). Inner electrodes sit at $z = \pm h$ and are grounded. Outer electrodes at $z = \pm(h + h_1)$ are shorted to the inners under short-circuit, and form a single unconstrained bus with $Q = 0$ under open-circuit.

Kirchhoff kinematics $u_z = w(r, \theta, t)$ in the host, with the usual plane-stress reduction of each piezo layer, produce the reduced constants

$$\bar{e}_{31} = e_{31} - \frac{c_{13}^E}{c_{33}^E} e_{33}, \quad \bar{\Xi}_{33} = \Xi_{33} + \frac{e_{33}^2}{c_{33}^E}. \quad (1)$$

For the PZT-4 values above, $\bar{e}_{31} = -13.05$ C m $^{-2}$: the correction $c_{13}^E e_{33} / c_{33}^E = 8.95$ C m $^{-2}$ is opposite in sign to the printed e_{31} and more than twice its size, so it dominates the reduced constant. Every electromechanical effect in this paper scales with $\bar{e}_{31}^2$, so the sign of the tabulated e_{31} matters: read as $+4.1$ it would give $\bar{e}_{31} = -4.85$ C m $^{-2}$ and couplings $(13.05/4.85)^2 = 7.24$ times smaller. Section 5 shows that Duan, Quek and Wang's Table 4 and independent finite elements both resolve that difference. Bending stiffness splits into a host contribution and a purely elastic contribution from the two layers,

$$d_1 = \frac{2}{3} \frac{E h^3}{1 - \nu^2}, \quad \bar{c}_{11} = c_{11}^E - \frac{(c_{13}^E)^2}{c_{33}^E}, \quad d_2 = \frac{2}{3} \bar{c}_{11} [(h + h_1)^3 - h^3]. \quad (2)$$

The electromechanical moment is proportional to $\bar{e}_{31}$ times the through-thickness electric field. The term d_2 is present whether the electrodes are open or shorted; it is ordinary added-layer stiffness, and it is what Duan, Quek and Wang's Fig. 3 isolates as the dominant short-circuit frequency shift.

3 Short-circuit determinant

Short-circuit electrodes on both faces of each layer are encoded by an interior potential that vanishes on both faces of the layer,

$$\varphi = \bar{\varphi}(r, \theta, t) \frac{4(z - h)(h + h_1 - z)}{h_1^2} \quad (3)$$

in the top layer, with its mirror below, so $\varphi = 0$ at $z = \pm h$ and at $z = \pm(h + h_1)$. The equation for $\bar{\varphi}$ is the stationarity condition of the thickness-integrated electric enthalpy, the same Galerkin projection that produces the flexural equation. Duan, Quek and Wang [5] used a sinusoidal profile, $\sin(\pi(z - h)/h_1)$, and closed its electrical equation instead by integrating Gauss's law through the layer; that closure is not conjugate to the moment through which the coupling reaches the flexural equation, and in the long-wave limit it overstates each layer's own short-circuit stiffening, $h_1^3 \bar{e}_{31}^2 / (12 \bar{\Xi}_{33})$, by exactly $12/\pi^2$. The quadratic (3) is the one-term profile whose projected long-wave stiffening is exact (Supplementary §S.1). On this steel-dominated stack the two closures differ by at most 0.011% in any Table 4 root, below the rounding of that table's integers (§5.1). Substituting (3) into the flexural operator and the projected electrical equation, and positing $\bar{\varphi} = \chi w$ with χ independent of r , produces two Helmholtz equations for w that are simultaneously satisfiable only when χ and the radial eigenvalue λ obey a cubic consistency condition (Supplementary §S.1). The cubic has three roots (χ_i, λ_i) , $i = 1, 2, 3$, each a function of the trial frequency ω . For each root the radial problem is second-order Helmholtz, with Bessel pair (J_n, Y_n) or modified pair (I_n, K_n) according to the sign of λ_i . Three branches times two radial functions is a six-dimensional basis, matching the sixth-order coupled system. The general solution is a sum over all three branches.

Edge conditions per radius are mechanical plus the insulated-edge condition $\partial\varphi/\partial r = 0$. The six rows of the short-circuit matrix, in order, are

$$\begin{array}{ll} \text{clamped-clamped:} & w, w', \partial\varphi/\partial r \quad \text{at } r = r_i \text{ and } r = r_o, \\ \text{free-free:} & M_{rr}, Q_r^{\text{eff}}, \partial\varphi/\partial r \quad \text{at } r = r_i \text{ and } r = r_o. \end{array} \quad (4)$$

Here $Q_r^{\text{eff}} = Q_r + r^{-1} \partial M_{r\theta} / \partial \theta$ is the Kirchhoff effective shear. The twisting moment $M_{r\theta}$ carries no piezoelectric term, and the correction it adds to Q_r is proportional to n^2 , so $Q_r^{\text{eff}} = Q_r$ at $n = 0$. Either pair is a 6×6 determinant. Zeros are found by a real-part sign-flip bisection on a pre-registered frequency bracket, with a negative-control bracket that is not allowed to flip. Published frequencies never seed a search; they enter only when a converged root is scored against them.

Setting $\bar{e}_{31} \rightarrow 0$ is a singular limit of the cubic (the leading coefficient and two subleading coefficients vanish together), so the elastic bilayer $d_1 + d_2$ is assembled as a separate two-branch 4×4 , not by zeroing $\bar{e}_{31}$ inside the 6×6 . That 4×4 is the short-circuit operator of the linear-potential model in §4, because $V = 0$ implies $E_z = 0$ and $M_p = 0$.

4 Open-circuit determinant

Open-circuit of the same fully electroded stack is not a flag on the 6×6 . Inner electrodes (the host/piezo interfaces) remain grounded, because the host is a conductor. Outer electrodes form a conducting bus at unknown voltage $V(t)$, spatially constant, with total charge $Q = 0$. The leading through-thickness field is then linear,

$$\varphi_{\text{top}} = V \frac{z - h}{h_1}, \quad \varphi_{\text{bottom}} = -V \frac{z + h}{h_1}, \quad (5)$$

for a same-poling parallel bimorph. Laboratory E_z is opposite in the two layers, extra in-plane stresses are opposite, and the moments *add*:

$$M_p = 2\bar{e}_{31}(h + h_1/2)V, \quad (6)$$

spatially constant. A constant isotropic moment does not enter Q_r^{eff} or the interior plate operator. It enters only the natural condition $M_{rr} = 0$.

The bus charge on both outers is the variable conjugate to V in the same electric enthalpy,

$$Q = 2\bar{e}_{31}(h + h_1/2) \int_A \nabla^2 w \, dA + C_0 V, \quad C_0 = \frac{2\bar{\Xi}_{33}A}{h_1}, \quad A = \pi(r_o^2 - r_i^2). \quad (7)$$

Its motional arm $h + h_1/2$ is the layer mid-surface, the same arm as M_p in (6), so the open-circuit operator is reciprocal; in the exact thin-plate solution D_z is uniform through each layer and couples only to the layer's mean strain. Evaluating the linear field's D_z at the outer electrode instead gives the arm $h + h_1$, which overstates every open-circuit split below by $(h + h_1)/(h + h_1/2)$ (7.7%, 11%, and 17% at the three Table 4 ratios). The interior potential (3) contributes no bus charge, because its field integrates to zero across each layer. Green's identity on the annulus gives $\int_A \nabla^2 w \, dA = 2\pi S$ at $n = 0$, with $S = r_o w'(r_o) - r_i w'(r_i)$, and vanishes at $n \neq 0$ by the θ -integral. Open-circuit $Q = 0$ therefore determines V from S when $n = 0$, and is vacuous when $n \neq 0$. Substituting into $M_{rr} = 0$ produces a rank-1 update of the elastic free-free 4×4 : both M_{rr} rows receive αS ,

$$\alpha = \frac{4\bar{e}_{31}^2(h + h_1/2)^2 h_1}{\bar{\Xi}_{33}(r_o^2 - r_i^2)}, \quad (8)$$

the Q_r^{eff} rows are unchanged, and the sign $+\alpha$ is the one that raises the $n = 0$ root. For $n \neq 0$ or $h_1 = 0$ the update is identically zero. That is not a numerical observation: at $h_1/2h = 1/12$ the first free-free $n = 1$ root of the open-circuit 4×4 is bit-identical to the elastic bilayer, 1756.00 rad s⁻¹. A continuous electrode cannot collect net charge from a nodal-diameter mode.

4.1 Clamped-clamped open-circuit is short-circuit

If $w' = 0$ on the entire boundary, Green's identity forces $\int_A \nabla^2 w \, dA = 0$. Equation (7) collapses to $Q = C_0 V$, so $Q = 0$ implies $V = 0$. Clamped-clamped open-circuit coincides with short-circuit, and with the elastic bilayer, at this order. The only remaining clamped-clamped coupling is the interior potential already in the 6×6 , whose frequency effect is 0.01% in Duan, Quek and Wang's Fig. 3 [5]. A frequency-based inverse for e_{31} from a clamped-clamped short-circuit mode is therefore asking a 0.01% correction to identify a constitutive constant. Section 6 measures that as a 330% uncertainty at 0.01% frequency precision. Switching the same mode to open-circuit does not help. Free-free axisymmetric motion does: $S \neq 0$ in general, V is alive, and M_p is seen by $M_{rr} = 0$. A mixed mechanical edge is the intermediate case. Clamped inner and free outer (C-F; Paper 3's inner-then-outer naming) forces $w'(r_i) = 0$, so $S = r_o w'(r_o)$ on the mode and the rank-1 update lives only on the outer M_{rr} row. Free inner and clamped outer (F-C) is the swap. Neither is a coincidence theorem: one edge remains free, so S need not vanish.

5 Forward validation

5.1 Clamped-clamped short-circuit against Table 4

Table 1 reports every non-trivial point of Duan, Quek and Wang's Table 4 [5]: circumferential orders $p = 0, 1, 2$, the first three radial indices, and the three thickness ratios. Targets are the

printed integers. The 6×6 is searched from scratch; a pre-registered relative bar of 0.15% is applied after the fact and is not used to retune a root. All 27 points pass, and every root rounds to the printed integer (largest absolute difference 0.49 rad s^{-1}). The worst relative error is -0.016% , at $p = 0$, first radial, $h_1/2h = 1/8$, which is within the rounding of that entry. Duan, Quek and Wang’s own sine closure also passes the bar at every point (worst $+0.013\%$) but rounds to only 19 of the 27 integers; no root differs between the two closures by more than 0.011% . The table is sensitive to the sign of e_{31} . With $e_{31} = +4.1 \text{ C m}^{-2}$ the same operator falls systematically below the table, by up to 0.049% at $h_1/2h = 1/5$, with a bias that grows with layer thickness, as a short-circuit stiffening 7.24 times too weak would. The printed sign removes it. The integer table still does not resolve the open-circuit splitting of §5.2, which is not a short-circuit quantity; this comparison confirms the layered Kirchhoff operator, its added-layer stiffness, and the short-circuit interior-potential stiffening.

Table 1: Clamped–clamped short-circuit frequencies (rad s^{-1}) against Duan, Quek and Wang Table 4 [5]. p is circumferential order; s is radial index as in that table. Bar: 0.15%. All 27 pass, and all 27 round to the printed integer.

p	s	$h_1/2h$	Table 4	present	rel. %
0	0	1/12	2792	2791.90	−0.0036
0	0	1/8	2853	2852.54	−0.0161
0	0	1/5	2989	2988.95	−0.0017
0	1	1/12	7723	7723.02	+0.0002
0	1	1/8	7891	7890.77	−0.0029
0	1	1/5	8268	8268.10	+0.0012
0	2	1/12	15182	15181.74	−0.0017
0	2	1/8	15512	15511.51	−0.0032
0	2	1/5	16253	16253.26	+0.0016
1	0	1/12	2928	2928.34	+0.0116
1	0	1/8	2992	2991.95	−0.0018
1	0	1/5	3135	3135.02	+0.0006
1	1	1/12	7965	7964.69	−0.0039
1	1	1/8	8138	8137.69	−0.0037
1	1	1/5	8527	8526.84	−0.0019
1	2	1/12	15482	15481.52	−0.0031
1	2	1/8	15818	15817.80	−0.0013
1	2	1/5	16574	16574.19	+0.0012
2	0	1/12	3477	3477.31	+0.0089
2	0	1/8	3553	3552.84	−0.0045
2	0	1/5	3723	3722.74	−0.0071
2	1	1/12	8768	8768.40	+0.0046
2	1	1/8	8959	8958.86	−0.0016
2	1	1/5	9387	9387.27	+0.0029
2	2	1/12	16434	16434.35	+0.0021
2	2	1/8	16791	16791.32	+0.0019
2	2	1/5	17594	17594.27	+0.0016

5.2 Free–free open-circuit, three thickness ratios

No literature table exists for free–free piezoelectric rings. Closed form is still exact. Table 2 reports the first $n = 0$ flexural root of the elastic bilayer, of the short-circuit 6×6 , and of the open-circuit rank-1 4×4 , at all three of Duan, Quek and Wang’s Table 4 thickness ratios. Short-circuit coupling

on this mode is +0.0052% to +0.053% relative to the bilayer: that is Duan, Quek and Wang’s Fig. 3 in numbers. Open-circuit coupling is 76 to 374 times larger, and it grows with $h_1/2h$: +1.950%, +2.776%, +4.016%. Thicker layers put more of the bending stiffness in the piezo and give the rank-1 moment a longer lever arm, so the identifying split is not a property of one geometry. A closed-form sweep of eleven ratios from $h_1/2h = 1/40$ to $1/4$ places those three of Duan, Quek and Wang’s Table 4 points on a smooth curve (Supplementary Fig. S.2). As $h_1/2h \rightarrow 0$ the split is linear in layer thickness: $\Delta_{OC}/(h_1/2h) \rightarrow 0.25$. The same sweep turns the three-point open-circuit e_{31} column of Table 4 into a curve: 0.01% frequency precision maps to 2.60% of e_{31} at $1/40$ and 0.40% at $1/4$. The first $n = 0$ radial overtone still splits, but more weakly: 3504.95 vs 3536.76 rad s^{-1} (+0.908%, $k_{\text{eff}}^2 = 0.0179$), and 0.01% frequency precision maps to 1.80% of e_{31} . The next sits at 8515.06 vs 8524.53 rad s^{-1} (+0.111%, 14.5% of e_{31} at 0.01% frequency). The identifying observable remains the fundamental.

Table 2: First free-free $n = 0$ flexural frequency. Closed-form values in rad s^{-1} (Hz in parentheses). Δ_{OC} is the rank-1 4×4 relative to the elastic bilayer. Short-circuit 6×6 coupling on this mode is +0.0052% to +0.053% and is omitted from the last column.

$h_1/2h$	bilayer	SC 6×6	OC 4×4	Δ_{OC}
1/12	747.986 (119.046)	748.025 (119.052)	762.569 (121.367)	+1.950%
1/8	764.113 (121.612)	764.234 (121.632)	785.327 (124.989)	+2.776%
1/5	800.303 (127.372)	800.726 (127.439)	832.439 (132.487)	+4.016%

External confirmation at $h_1/2h = 1/12$ is axisymmetric coupled-field finite elements on one mesh (PLANE223, KEYOPT(1)=1001), in the millimetre-newton-second-tonne system with relative permittivity. Coupling-off ($e_{ij} = 0$) gives 118.810 Hz. Both-faces short-circuit (VOLT= 0 on $z = \pm h$ and $z = \pm(h + h_1)$) gives 118.816 Hz, +0.0052% above coupling-off, against the closed-form short-circuit stiffening of +0.0052% in Table 2: true short-circuit on this mode is the bilayer plus the interior-potential correction, and nothing else. Open-circuit (VOLT= 0 on the interfaces; one coupled VOLT set on both outer faces, unconstrained) gives 121.126 Hz. The finite-element split relative to true short-circuit is +1.944%, against the closed-form +1.950% (ratio 0.997). The finite-element open circuit carries the interior potential as well as the bus field, so the like-for-like closed-form quantity is the open-circuit root with the interior potential added (§8) over the short-circuit 6×6 ; that is +1.949%, within 0.02% of the tabulated Δ_{OC} , which is why the table’s column is the one scored. Both modes are bending: transverse displacement has the same sign through the thickness at $r = r_o$, radial displacement is opposite, and the ratio of those components is 19.71 (short-circuit) and 20.06 (open-circuit). A deck that grounds only the outer faces and leaves the interface free sits at 121.859 Hz and is a mixed electrical condition, 2.56% above true short-circuit and above open-circuit as well; it is not used as the short-circuit baseline (Supplementary §S.3).

The same fair pair, cloned without remeshing to the other two Table 4 ratios, confirms the same stiffening trend. At $h_1/2h = 1/8$, open-circuit gives 124.722 Hz against 121.364 Hz both-faces short-circuit, a +2.766% split against the closed-form +2.776%; at $h_1/2h = 1/5$, 132.183 Hz against 127.105 Hz, a +3.995% split against the closed-form +4.016%. Every deck shows the same bending signature (ratio of thickness-through components 19.7–20.4) and a 0 Hz rigid-body mode, and no run printed an element-level error. The finite-element split is 99.7%, 99.6%, and 99.5% of the closed-form split. The same decks with $e_{31} = +4.1 \text{ C m}^{-2}$ give splits 6.7 to 7.0 times smaller (+0.280%, +0.405%, +0.600%), within 0.2% of the closed form at that sign, so the finite elements follow the $\bar{e}_{31}^2$ scaling of (8) rather than one calibrated point. An earlier version of this model took the bus charge from D_z at the outer electrode (arm $h + h_1$); finite elements then recovered (at $e_{31} = +4.1$) 93%, 90%, and 86% of its splits, which is $(h + h_1/2)/(h + h_1)$ at each ratio to within

0.5% and is how the non-reciprocal charge row was found. Doubling the piezo through-thickness divisions on the same electrical pair leaves the three finite-element splits unchanged at the printed precision.

Three-dimensional check at $n \geq 1$. The finite-element models above are axisymmetric, so they see $n = 0$ only. A three-dimensional coupled-field model (SOLID226 half ring, $0 \leq \theta \leq \pi$ with a symmetric cut, so every $\cos n\theta$ order appears in one solve; same materials, units, and electrical conditions) was run at $h_1/2h = 1/12$ and $1/5$ on two meshes (Supplementary §S.9). Its $n = 0$ roots reproduce the axisymmetric ones: coupling-off 118.8105 against 118.8102 Hz, and open-/short-circuit splits of 1.945% and 3.997% against 1.944% and 3.995%. At $n = 1-4$ it gives three results. Open circuit coincides with short circuit to 3×10^{-10} relative, with a bus voltage below 10^{-10} of its $n = 0$ value, as the no-net-charge argument of §4 requires. The short-circuit stiffening from the interior potential, 1.0×10^{-5} to 5.3×10^{-4} relative and non-monotone in n , is reproduced to within 2.6% of the 6×6 value at every $n = 0-4$ and both ratios; that is the coupled $n \geq 1$ content of the model, and nothing else checks it. The elastic roots themselves sit 0.53% to 1.19% below the thin-plate values at $n = 1-4$, against 0.20% and 0.26% at $n = 0$: one-signed, mesh-converged to 0.03%, largest at $n = 1$, and larger for the thicker layer at every n – the same pattern as the free-free shell-element comparison of [12]. It is a thickness effect, not an error in the $n \geq 1$ operator: on a bare steel annulus of the same planform at 20, 10 and 5 mm, with the mesh error removed by Richardson extrapolation over three uniformly refined meshes, the $n = 1-4$ offset falls from 0.49–0.92% to 0.11–0.16%, and its extrapolation to zero thickness is within 0.015% of zero at every $n = 0-4$ (Supplementary §S.9). The $n = 0$ offset is purely quadratic in thickness; at $n \geq 1$ a first-order term of comparable or larger size appears, as expected of the free-edge boundary layer that a Kirchhoff effective-shear edge omits. The superseded Q_r free-edge row put $n = 1$ at $h_1/2h = 1/12$ at $1573.49 \text{ rad s}^{-1}$; the three-dimensional root is 10.5% above it, and 1.0% below the effective-shear value.

5.3 Clamped-free and free-clamped open-circuit

The elastic mixed-edge 4×4 is Paper 3's construction [12]: C-F replaces the inner M_{rr} and Q_r^{eff} rows by $w = w' = 0$; F-C is the swap. At $h_1 = 0$ those determinants recover the live Paper 3 ring solver on this geometry to 4.5×10^{-10} (C-F) and 2.3×10^{-10} (F-C) relative. Short-circuit coupling on the mixed 6×6 remains the usual interior-potential correction, +0.0061% (C-F) and +0.0039% (F-C) at $h_1/2h = 1/12$. Open-circuit does not coincide with that bilayer. Table 3 records the first $n = 0$ root. C-F splits by +1.826% ($k_{\text{eff}}^2 = 0.0355$), comparable to the free-free +1.950% at the same ratio, because the free outer radius still carries $S = r_o w'(r_o)$. F-C splits by +0.039% ($k_{\text{eff}}^2 = 0.00078$): only the inner hole is free, so $S = -r_i w'(r_i)$ and the rank-1 moment acts on the smaller edge. $n \neq 0$ open-circuit is bit-identical to the mixed elastic 4×4 on both pairs, as for free-free. Axisymmetric finite elements on the same mesh as §5.2, with only the inner radius clamped ($U_X = U_Y = 0$ through the stack, outer free), confirm the C-F split: both-faces short-circuit 67.076 Hz against open-circuit 68.290 Hz, +1.811%, against the closed-form +1.826% (ratio 0.992). No 0 Hz rigid body (the clamp took). Both modes are bending (U_Y same sign through the thickness at $r = r_o$, U_X opposite, ratio 45.5 and 47.0). F-C was not meshed.

Table 3: First $n = 0$ mixed-edge flexural frequency at $h_1/2h = 1/12$, closed form, rad s^{-1} (Hz). C-F is inner clamped, outer free; F-C is the swap. Δ_{OC} is relative to the mixed-edge elastic bilayer.

edge	bilayer	SC 6×6	OC 4×4	Δ_{OC}
C-F	421.895 (67.147)	421.920 (67.151)	429.599 (68.373)	+1.826%
F-C	899.061 (143.090)	899.096 (143.096)	899.410 (143.146)	+0.0388%

6 Inverse recovery

The unknown is a single material constant p ; every other constant is held at its true value. A target ω_* is the exact forward root at p_{true} . The residual $G(p) = \omega_{\text{root}}(p) - \omega_*$ is bisected on a bracket that contains a genuine sign flip; a hi-side-only bracket is the negative control. On-branch finite differences give $dp/d\omega$, hence the uncertainty in p at an assumed relative precision in ω . The same recipe inverted Southwell’s b/a at fixed λ^2 for the elastic ring [12]. Round-trip recovery of p_{true} from ω_* is 4×10^{-9} relative against a 10^{-6} bar in every case below; the inverse machinery is not the issue. The scientific content is the propagated uncertainty.

Table 4 records host E from the clamped-clamped short-circuit fundamental at $h_1/2h = 1/12$, and e_{31} from that same short-circuit family and from the free-free and clamped-free open-circuit $n = 0$ roots, at all three thickness ratios for free-free and at $h_1/2h = 1/12$ for clamped-free. Host E is well conditioned under short-circuit, as required by $\omega \sim \sqrt{E}$ (relative E uncertainty ≈ 2 times frequency precision). Short-circuit e_{31} is not: 0.01% in frequency becomes 330%, 109%, and 33% in e_{31} as the layer thickens. Open-circuit reduces those figures to 0.87%, 0.63%, and 0.45%. Clamped-free open-circuit at $h_1/2h = 1/12$ is 0.93%, the same class as free-free. Thickness helps both electrical conditions, because more of the bending stiffness sits in the piezo, but it does not close the electrical gap: at $h_1/2h = 1/5$ short-circuit is still 72 times worse than open-circuit. Figure 2 is the same comparison.

Those open-circuit figures hold E at its true value, which no experiment can do. The last row of Table 4 instead recovers E from the clamped-clamped short-circuit fundamental first, then e_{31} from the free-free open-circuit root using that recovered $\hat{E}$, and propagates $\hat{E}$ ’s own uncertainty forward into e_{31} – the two-step protocol an actual measurement would follow, at $h_1/2h = 1/12$. The result is 1.21% of e_{31} at 0.01% frequency precision: a direct term of 0.87%, matching the idealized row above, plus a nearly equal term of 0.84% from propagating $\hat{E}$ ’s own 0.025% uncertainty through the open-circuit root’s weak but nonzero dependence on E , combined in quadrature. Substituting the clamped-clamped open-circuit root for the free-free one in the second step – the negative control – reproduces the 330% short-circuit figure instead (330.3%), confirming that the sequential protocol still requires the free-free open-circuit observable and gains nothing from the clamped-clamped one. Host E recovered from free-free $n = 0$ elastic or open-circuit flexure is the same thermometer as the clamped-clamped short-circuit channel (0.025% and 0.026% at 0.01% frequency). From the same open-circuit fundamental, host ν and the piezo-layer C_{11}^E recover at 0.118% and 0.074%; Ξ_{33} from that frequency is 0.71%, while the clamped capacitance C_0 identifies Ξ_{33} at roundoff. Frequency constrains Ξ_{33} only as well as it constrains e_{31} , because both enter through $\bar{e}_{31}^2/\bar{\Xi}_{33}$; C_0 is the dielectric thermometer.

Table 4: Single-parameter inverse. Round-trip relative error on p_{true} is 4×10^{-9} in every $h_1/2h = 1/12$ row and 1.6×10^{-7} in the sequential row (bar 10^{-6}); the $1/8$ and $1/5$ rows are on-branch sensitivities. Propagated relative uncertainty is at 0.01% frequency precision. C–C SC: clamped–clamped short-circuit fundamental. F–F OC: free–free open-circuit, $n = 0$. C–F OC: clamped–inner, free–outer open-circuit, $n = 0$. The three E rows are independent channels of the same $\omega \sim \sqrt{E}$ thermometer. The last row recovers E from the first row, then e_{31} from the F–F OC e_{31} row, propagating $\hat{E}$ ’s uncertainty forward.

forward model	unknown	unc. at 0.01% freq.
C–C SC, $h_1/2h = 1/12$	$E = 2 \times 10^{11}$ Pa	0.025% of E
F–F elastic, $h_1/2h = 1/12$	E	0.025% of E
F–F OC, $h_1/2h = 1/12$	E	0.026% of E
C–C SC, $h_1/2h = 1/12$	e_{31}	330% of e_{31}
C–C SC, $h_1/2h = 1/8$	e_{31}	109% of e_{31}
C–C SC, $h_1/2h = 1/5$	e_{31}	33% of e_{31}
F–F OC, $h_1/2h = 1/12$	e_{31}	0.87% of e_{31}
C–F OC, $h_1/2h = 1/12$	e_{31}	0.93% of e_{31}
F–F OC, $h_1/2h = 1/8$	e_{31}	0.63% of e_{31}
F–F OC, $h_1/2h = 1/5$	e_{31}	0.45% of e_{31}
F–F OC (seq. $\hat{E}$), $h_1/2h = 1/12$	e_{31}	1.21% of e_{31}

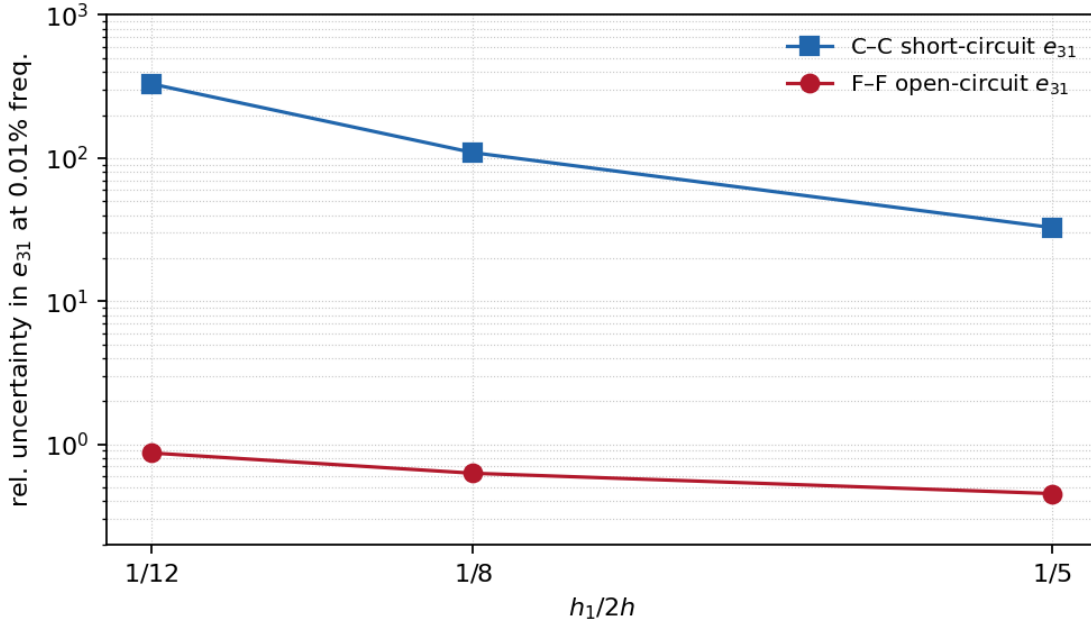


Figure 2: Propagated relative uncertainty in e_{31} at 0.01% frequency precision, on the clamped–clamped short-circuit fundamental and on the free–free open-circuit $n = 0$ root. Both improve as $h_1/2h$ grows; the electrical gap does not close.

Two caveats sit on the open-circuit inverse. First, ω_{OC} depends on $\bar{e}_{31}^2$, so a frequency determines $|\bar{e}_{31}|$ and therefore two raw e_{31} values reflected about $(c_{13}^E/c_{33}^E)e_{33} \approx 8.95 \text{ C m}^{-2}$. Duan, Quek and Wang’s $e_{31} = -4.1$ is the lower branch; the twin is near 22.0 C m^{-2} , outside the bracket $[-10, -1] \text{ C m}^{-2}$ used here. Holding e_{33} known, as a single-parameter recovery requires, selects the branch. A joint (e_{31}, e_{33}) fit would re-open the reflection. Second, a joint (E, e_{31}) fit from one short-circuit frequency would be worse than the e_{31} -only short-circuit row, not better: one well-

conditioned parameter cannot regularize an unidentifiable one. Additional short-circuit flexural modes of the same family will not collapse the 330% figure, because they share the same 10^{-4} -level coupling. Open-circuit on a clamped-clamped mode will not either, by §4.1. The identifying observable for e_{31} is an axisymmetric open-circuit frequency on a plate that has at least one free edge: the free-free $n = 0$ root, or the C-F $n = 0$ root of Table 3 (0.93% of e_{31} at 0.01% frequency, against free-free's 0.87%). F-C open-circuit is not: its 0.039% split is 47 times smaller, and is a result, not a failed search.

The 0.87%/0.93% figures above assume 0.01% frequency precision, an optimistic narrow-band-tracking number rather than a measured instrument comparison. Figure 3 extends the free-free and clamped-free open-circuit inverse along the frequency-precision axis itself, rather than reporting only the two columns of Table 4: the propagated relative uncertainty in e_{31} is linear in the assumed relative frequency error over three decades, by construction of the linear-propagation formula, so there is no threshold below which the recovery qualitatively changes and no floor at which it saturates. A lab-realistic floor, taken as the impedance-analyzer-versus-scanning-laser-Doppler-vibrometer frequency disagreement measured on a physically distinct PZT rod specimen in companion experimental work from this group [13] (mean 0.10% across five axial modes), lands within a few percent of the already-tabulated 0.1% column for both channels (8.9% and 9.5% of e_{31} , against 8.7% and 9.3%). The gap between the idealized 0.01% precision and this cross-instrument comparison, not the choice between the free-free and clamped-free observables (which agree with each other to within about 7% at every precision tested), is the dominant source of uncertainty in a real inverse recovery.

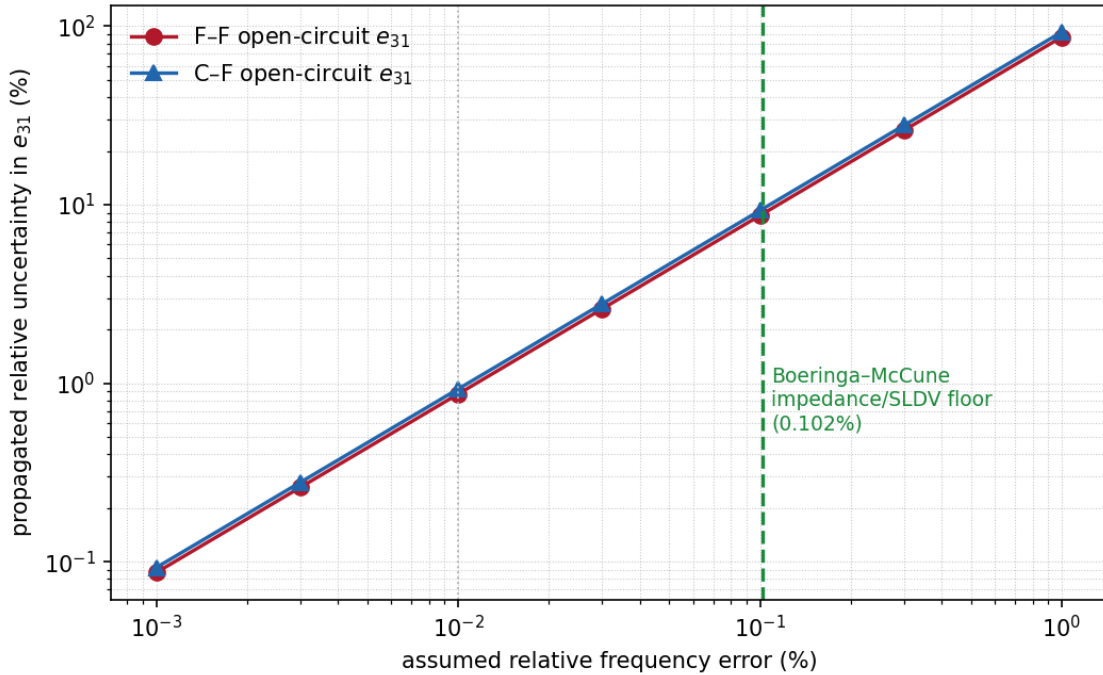


Figure 3: Propagated relative uncertainty in e_{31} versus assumed relative frequency error, free-free and clamped-free $n = 0$ open-circuit. Linear on log-log axes by construction of the propagation formula. The dashed line marks the impedance-analyzer/SLDV frequency-disagreement floor measured on a physically distinct rod specimen [13], which coincides closely with the already-tabulated 0.1% column.

Read as a measurement promise, 0.87% (F-F) and 0.93% (C-F) at 0.01% frequency precision

would require resolving these resonances to one part in ten thousand – finer than the impedance-analyzer/SLDV comparison of Fig. 3 delivers on a physically distinct rod. Read instead as a statement about well-posedness, the result stands regardless of instrument: $d\omega_{OC}/de_{31}$ is nonzero and large enough that *any* finite frequency measurement maps to a finite e_{31} interval, in contrast to the short-circuit clamped–clamped channel, where the same derivative is so small that no realistic frequency precision recovers e_{31} at all (330% at 0.01%, worse at any coarser floor). The 0.01% column is therefore best read as characterizing the conditioning of the inverse problem – how tightly a well-resolved frequency constrains e_{31} – rather than as a laboratory accuracy target; the number an actual bench measurement on this ring would deliver is closer to the Boeringa–McCune floor, roughly 9%. Both readings support the same qualitative conclusion, that the open-circuit observable unlocks e_{31} where short-circuit cannot; only the second is the number to quote for what a real experiment on such a ring would recover.

7 Driven admittance

Imposing V rather than $Q = 0$ converts §4 into a driven problem: the elastic 4×4 is solved with M_{rr} -row right-hand side M_p at both edges, and Q follows from (7). The undamped admittance is $Y(\omega) = j\omega Q/V$, or equivalently the real effective capacitance $C_{\text{eff}}(\omega) = Q/V$. For $n \neq 0$, $S = 0$ and $C_{\text{eff}} = C_0$ identically: a continuous electrode carries no net motional charge in a nodal-diameter mode. For $n = 0$, C_{eff} has a pole at the short-circuit (elastic) root and a zero at the open-circuit root. An independent bisection of Q/V at $h_1/2h = 1/12$ recovers $762.5693785054 \text{ rad s}^{-1}$ at a relative difference of 9×10^{-15} from the free-vibration 4×4 . The IEEE coupling factor [7] built from those two frequencies is

$$k_{\text{eff}}^2 = 1 - (f_r/f_a)^2. \quad (9)$$

At $h_1/2h = 1/12$, $1/8$, and $1/5$ this is 0.0379, 0.0533, and 0.0757 ($k_{\text{eff}} = 0.195, 0.231, 0.275$), tracking the open-circuit splits of Table 2. Independent bisection of Q/V at $1/8$ and $1/5$ recovers the matching open-circuit roots to 1.7×10^{-14} and 3×10^{-16} relative, so those k_{eff} values now come from the driven problem, not only from free vibration (Supplementary Fig. S.3). Clamped capacitance scales as $1/h_1$: $C_0 = 9.99 \mu\text{F}$ at $1/12$, $6.66 \mu\text{F}$ at $1/8$, and $4.16 \mu\text{F}$ at $1/5$. Off resonance at $1/12$, $C_{\text{eff}}/C_0 = 1.092$ at 400 rad s^{-1} (above unity, as a free plate must be) and 0.752 at 800 rad s^{-1} (just above the zero, returning to C_0 from below), which is the Butterworth–van Dyke shape with motional capacitance $C_m = C_0 k_{\text{eff}}^2 / (1 - k_{\text{eff}}^2)$. Figure 4 zooms on the 1.95% splitting; the same curve on a 95–145 Hz axis is Supplementary Fig. S.1. A harmonic PLANE223 run on the same mesh, with 1 V on the outer bus, places the pole at 118.82 Hz and the charge-zero at 121.126 Hz, against the modal both-faces-short-circuit / open-circuit pair 118.816 / 121.126 Hz on that mesh (analytic 119.052 / 121.367 Hz; both finite-element frequencies sit 0.20% below their analytic partners, the elastic Kirchhoff-versus-2D offset of the coupling-off deck). The driven finite-element pole/zero is the modal pair, not a third physics.

8 Discussion

The numerical inverse never failed. Every round-trip recovered the true constant to nine figures. What failed, under short-circuit flexure, is identifiability of e_{31} , and it failed for a reason that is in Duan, Quek and Wang’s Fig. 3 and in Green’s identity, not in the root search. Treating that failure as a finding, rather than as a defect to be tuned away, is what makes the open-circuit $n = 0$ result a result: it is the smallest change of electrical and mechanical boundary conditions that puts e_{31} on a frequency.

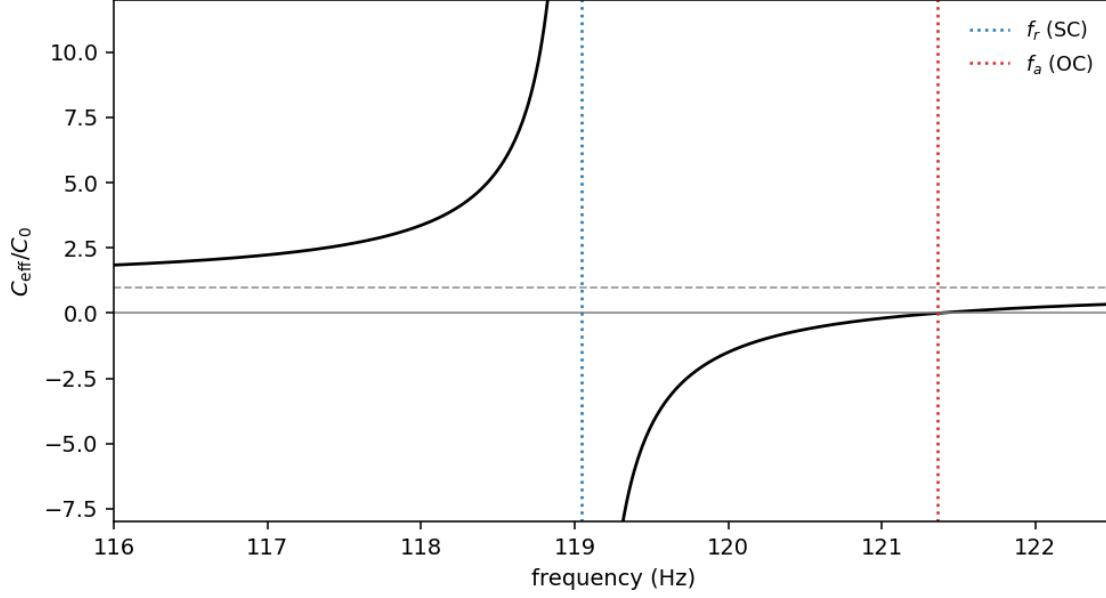


Figure 4: Effective capacitance of the driven free-free $n = 0$ model at $h_1/2h = 1/12$, $C_0 = 9.99 \mu\text{F}$. The pole is the elastic bilayer root $f_r = 119.046 \text{ Hz}$, the short-circuit root of the linear-potential model (the 6×6 puts it at 119.052 Hz); the zero is the open-circuit root $f_a = 121.367 \text{ Hz}$.

Limits, so they are not rediscovered as objections. The open-circuit model is leading-order linear potential; the short-circuit 6×6 retains the interior potential. Their coupling corrections differ by 76 to 374 times on the first $n = 0$ mode (1.95% versus 0.0052% at $h_1/2h = 1/12$), so comparing them as f_a and f_r is consistent at the precision of (9). This was checked, not only argued from magnitude: enriching the open-circuit potential with the same interior potential used by the short-circuit 6×6 raises the free-free $n = 0$ open-circuit root by 4.8×10^{-5} , 1.4×10^{-4} and 4.6×10^{-4} relative at the three ratios, 86–93% of the short-circuit interior stiffening. Under the consistent projection the uniform bus field and the interior field are orthogonal in the enthalpy, so the two stiffenings add at leading order, and the enriched open-circuit root over the short-circuit 6×6 reproduces Δ_{OC} of Table 2 to within 0.2% of its value at every ratio. The leading-order model used throughout is adequate here. Circumferential orders $n \neq 0$ carry no net charge on a continuous electrode and do not split; three-dimensional finite elements confirm this at $n = 1$ –4 to 10^{-10} . Clamped-clamped open-circuit does not split at this order. Clamped-free $n = 0$ open-circuit does, at nearly the free-free split; free-clamped $n = 0$ open-circuit splits far less, because only the inner hole is free. In-plane piezoelectric coupling, a monolithic ring, and simultaneous multi-parameter recovery are not claimed. No laboratory specimen is reported here. Were one measured, the free-free $n = 0$ resonance/antiresonance pair of Table 4 is the observable; the 0.01% column of that table characterizes inverse conditioning, not a laboratory accuracy target (Fig. 3).

The elastic ring of McCune and Stutts [12] is the mechanical baseline this construction sits on. The present frequencies do not re-score Vogel, Narita, or Irie; they score a piezoelectric table and an open-circuit splitting that that elastic paper did not contain.

The solver implementing the 6×6 and the rank-1 4×4 , and the scripts that produced every table here, are in the public tree accompanying McCune and Stutts [12].

9 Conclusions

A layered piezoelectric annulus has a short-circuit 6×6 that recovers Duan, Quek and Wang’s Table 4 at 27/27, every root rounding to the printed integer (worst relative error -0.016%), and a free–free open-circuit 4×4 whose first $n = 0$ root lies 1.950%, 2.776%, and 4.016% above the elastic bilayer at $h_1/2h = 1/12$, $1/8$, and $1/5$. Piezoelectric finite elements at all three ratios reproduce that split against both-faces short-circuit to 99.5–99.7% of its value, once the bus charge is taken as the variable conjugate to the bus voltage, and reproduce the 6.7- to 7.0-fold smaller split that a +4.1 reading of e_{31} would give to within 0.2%. Clamped–free open-circuit on the same mesh reproduces its closed-form +1.826% split to within 0.9%. Three-dimensional coupled-field elements extend the check to $n = 1$ –4: open circuit equals short circuit there, and the short-circuit interior-potential stiffening is reproduced to within 2.6%. Host E is recoverable from one short-circuit clamped–clamped frequency at 0.025% for 0.01% frequency precision, and from free–free elastic or open-circuit flexure at 0.025% and 0.026%; e_{31} from the clamped–clamped short-circuit observable is not (330%). The same e_{31} is recoverable from the free–free axisymmetric open-circuit frequency at 0.87%, 0.63%, and 0.45% as the layer thickens, assuming E known exactly; short-circuit at the same three ratios remains 330%, 109%, and 33%. Recovering E and e_{31} in sequence, as an actual measurement would, compounds the $h_1/2h = 1/12$ open-circuit figure to 1.21% – still two orders of magnitude better than the short-circuit route. The driven admittance of the open-circuit model has its pole and zero at those two frequencies at all three Table 4 ratios, with $k_{\text{eff}} = 0.195$, 0.231, and 0.275. That pair, not a short-circuit flexural list, is the identifying observable for e_{31} . A clamped-inner, free-outer ring is a second channel of the same class (+1.826% closed-form, reproduced by finite elements to 0.9%, 0.93% of e_{31} at 0.01% frequency); the swapped pair is not.

Data and code availability

The solver implementing the operators of this paper, the scripts that produced every number and table here, and the finite-element decks with their text outputs extend the public tree accompanying [12].

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CRedit authorship contribution statement

Garrek McCune: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Daniel Stutts:** Conceptualization, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used generative AI tools to assist with manuscript and code organization, drafting, editing, and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

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